# ME 5309 – Mid-Semester Examination

Answer Booklet

**Instructions:** Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

## Question 1 [10 marks]

### Q1(a) [4 marks]

(i) Pixel pitch p = 5 microm = 0.005 mm. Hence:

$$f_x = f/p = 16/0.005 = 3200 pixels$$

(ii) Sensor width = 1280(0.005) = 6.40 mm. At standoff Z = 400 mm, belt width W = Z(6.40)/16 = 160 mm.

$$Scale = 160/1280 = 0.125 mm/pixel$$

*Working:* $5\,\mu$m $= 5 \times 10^{-3}$ mm $= 0.005$ mm. So $16/0.005 = 16 \times 200 = 3200$ pixels. For the sensor, $1280 \times 0.005 = 6.4$ mm. Then $W = (400/16) \times 6.4 = 25 \times 6.4 = 160$ mm, and $160/1280 = 0.125$ mm/pixel.

### Q1(b) [3 marks]

ADC step = 5/2^12 = 0.0012207 V = 1.2207 mV. Sensor sensitivity is 10 mV/N, so ΔF = 1.2207/10 = 0.12207 N. The smallest resolvable force change is approximately 0.122 N.

*Calculation:* $2^{12} = 4096$ levels. One step is $5/4096 = 0.0012207$ V, i.e. $1.2207$ mV. Dividing by the sensitivity, $1.2207\,\text{mV}/(10\,\text{mV/N}) = 0.12207$ N. The mV units cancel, leaving N.

### Q1(c) [3 marks]

$$F = e^( - 2 \times 0.1) = 0.818731$$

I used G = (1-F)/(-2) = -0.0906346. So:

$$x_{k+1} = 0.818731 x_k - 0.0906346 u_k$$

## Question 2 [10 marks]

### Q2(a) [2 marks]

At equilibrium the velocity should be zero. The torque and gravity torque must balance.

### Q2(b) [2 marks]

The Jacobian has $A_{12}$ = 1 and $A_{22}$ = -b/J = -2.5. The input matrix is B = [0, 50]$^T$.

### Q2(c) [4 marks]

At rest, 50(π/2-θss) = 2.4525. This gives θss = 1.5218 rad = 87.19°, so error approximately 2.81°.

### Q2(d) [2 marks]

Gravity compensation makes the steady-state error zero. It does not change the closed-loop poles because it only adds a constant torque.

## Question 3 [10 marks]

### Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

### Q3(b) [2 marks]

Tustin uses s = (2/T)(z-1)/(z+1).

### Q3(c) [4 marks]

The characteristic polynomial is $s^2+(\ell_1+2)s+(2\ell_1+\ell_2)$. I match it to $s^2+10s+100$ and get $\ell_1=8$, $\ell_2=84$.

### Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.

## Question 4 [10 marks]

### Q4(a) [2 marks]

Poles are -2±2j and Nff=8.

### Q4(b) [2 marks]

With actual b=0.75 and unchanged Nff=8, $k_1=2$, the steady-state relation is θss = bNff/(c+bk₁) = 0.75x8/(6+0.75x2)=6/7.5=0.8. For r=1, percentage error = (1-0.8)x100%=20%.

With the actual gain, the steady-state output is $y_{ss} = b_{actual}N_{ff}r/(c+b_{actual}k_1)$ for the stated setup. Substituting the given numbers gives $y_{ss} = 0.8$ for a unit reference. Therefore $e_{ss} = 1-0.8 = 0.2$, i.e. a 20% error.

## Q4(c) [4 marks]

With $\dot{\xi}$=r-θ and u=-$k_1$θ-$k_2\dot{\theta}$+K_Iξ, the homogeneous closed-loop characteristic polynomial is $s^3$+(a+b$k_2$)$s^2$+(c+
Desired polynomial: (s+4)[$(s+2)^2$+4]=(s+4)($s^2$+4s+8)=$s^3$+8$s^2$+24s+32. For a=2,b=1,c=6, matching coefficients gives $k_2$=6, $k_1$=18 and K_I=32.

First multiply $(s+4)(s^2+4s+8) = s^3+8s^2+24s+32$. Comparing this with the augmented characteristic polynomial gives $k_2 = 6$, $k_1 = 18$, and $K_I = 32$. I checked the coefficients in descending powers of $s$.

## Q4(d) [2 marks]

At steady state, a stable closed loop with integral action drives $\dot{\xi}$=r-θ to zero; therefore r-θ=0 and the tracking error is zero, assuming no saturation or other implementation limits prevent convergence. A practical failure mode is integrator windup during actuator saturation; the standard remedy is anti-windup, such as back-calculation or conditional integration.

With integral action, a constant tracking error keeps accumulating in the integral state, so the controller can adjust its output until the error goes to zero, assuming the closed loop is stable and the actuator is not constrained. Under saturation the integral state can keep growing; anti-windup limits or back-calculates that state.